

A Rigorous Behavior Assessment of CNNs Using a Data-Domain Sampling Regime

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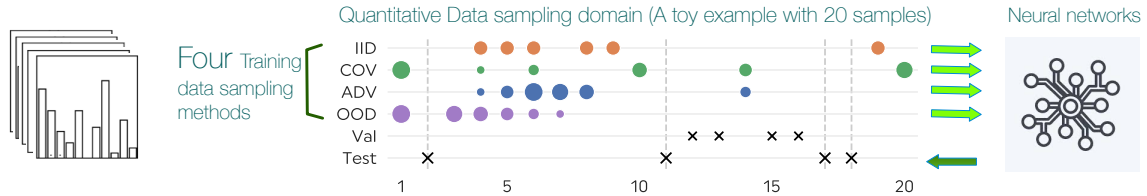


Figure 1: **A toy example of a training data sampling regime.** We quantify behaviors of convolutional neural networks (CNNs) in response to training inputs. Four distinct data-domain sampling methods are employed to supervise CNNs. These resulting CNNs are then evaluated on a same test set to assess their behavior differences.

Abstract—We present a data-domain sampling regime for quantifying CNNs’ graphic perception behaviors. This regime lets us evaluate CNNs’ ratio estimation ability in bar charts from three perspectives: *sensitivity to training-test distribution discrepancies*, *stability to limited samples*, and *relative expertise to human observers*. After analyzing 16 million trials from 800 CNN models and 6,825 trials from 113 human participants, we arrived at a simple and actionable conclusion: CNNs can outperform humans and their biases simply depend on the training-test distance. We show evidence of this simple, elegant behavior of the machines when they interpret visualization images. osf.io/gfqc3 provides registration, the code for our sampling regime, and experimental results.

Index Terms—Quantification, convolutional neural network, sampling, graphical perception, evaluation

1 INTRODUCTION

A crucial consideration when using machine learning models, such as a convolutional neural network (CNN) or a large-language model, for visualization tasks is understanding when to trust their results and how to effectively train them [26]. Assessing their behavior quantitatively is essential for evaluating their reliability and identifying potential limitations. While most quantification studies focus on algorithmic accuracy, appearance-based robustness (e.g., changes in background colors or line widths [13]), model vulnerabilities [20, 54], and lack of visual literacy [3, 19, 25, 52], it remains rare for evaluations to explicitly address the data-domain—the sampling of data attributes encoded in the visual format. Haehn et al. [20] examined the data domain (e.g., ratios in bar charts), but their training, test, and validation data samples were restricted to the ratios from Cleveland and McGill [8], which are essentially the *test* samples for *human* observers. To rigorously evaluate both unsupervised and supervised algorithms, the training data must be carefully defined, as its selection can give observers advantages or disadvantages when tested and compared [5].

Furthermore, the quantification of algorithmic behaviors on data-domain can facilitate alignment with human observers’ abilities. For example, as even for the most popular visualizations (e.g., bar charts and scatter plots), the ability to perform chart-reading tasks is considerably challenging for human observers who rely on the coordination of several mental processes, including rapid perceptual computations [8]

with respect to a visualization schema [43], ability to perform abstraction [41], working memory [42], domain knowledge [56], and interpretation processes [50]. Equally important is the recognition that the behavior constraints of CNNs differ from those of humans [16]. Thus principles and methods derived from human behaviors must be explicitly quantified for machine observers [24].

The goal of this work is to fill in the knowledge gaps by critically quantifying supervised discriminative CNNs by focusing on data-domain sampling. In particular, we study model behaviors by manipulating training-test data distribution shift to answer the following question: *How do sampled training data values influence the model behaviors on the test set?* Through multiple experiments, we assess algorithmic behaviors from several perspectives: models’ robustness to out-of-distribution data samples (*sensitivity*), resilience to limited input resources (*stability*), and when they surpass humans (*relative expertise*). Together, these perspectives provide a holistic view of a model’s capabilities in a visualization context.

Method. To systematically answer our research question and quantify CNNs’ behaviors in visualization contexts, we followed reliable tests used in human studies that focus on data targets of bar heights [2, 49] or ratios [8, 22]. We then designed a training data sampling regime to enable direct behavior quantification during the test phase, after training models from shifting training-test data sampling distributions (see a toy example in Figure 1): (1) **Independent and identical data distribution (IID)**: simple random sampling from the same data domain as the test set, as the baseline. (2) **Coverage (COV)**: sampling values that are distant from each other in the *training* set; (3) **Adversarial (ADV)**: sampling values that are positioned away from the *test* set values; (4) **Out-of-distribution (OOD)**: sampling values concentrated in the left portion of the data domain. Through three controlled and two exploratory studies, we systematically quantified the reading behavior changes in response to these training data samples, after evaluating models on the same test set in each run so that behaviors between sampling methods are comparable.

Results. Our work provides new findings that help interpret CNNs’ ability to decode charts. First, CNNs achieved higher accuracy and

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Received 31 March 2025; revised 1 July 2025; accepted 15 July 2025.

Date of publication 1 December 2025; date of current version 3 February 2026.

This article has supplementary downloadable material available at <https://doi.org/10.1109/TVCG.2025.3633829>, provided by the authors

Digital Object Identifier no. 10.1109/TVCG.2025.3633829

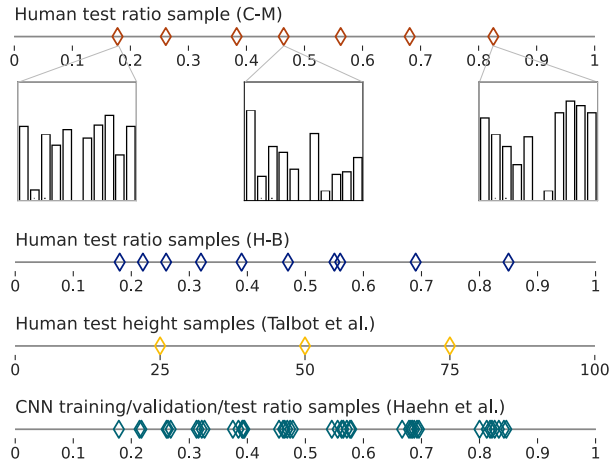


Figure 2: **Data sampling configurations in published studies.** Human studies (1st to 3rd rows): Ratio or height samples as seen by human observers in Cleveland and McGill (C-M) [8], Heer and Bostock (H-B) [22], and Talbot et al. [49]. An algorithmic study (Last row): Haehn et al. [20] reused test samples in the human experiments as the data-domain to sample the training, test, and validation sets for CNNs.

lower uncertainty with IID and COV in the ratio estimation task (section 5). Second, reducing the number of training samples increases both inference error and uncertainty, which explains the underperforming CNNs in Haehn et al.’s experiment [20]. Third, an unexpected finding is the benefits of the coverage sampling COV: increasing training data span and evenly distributing samples within the span leads to higher stability than the most widely used IID-supervised models, especially when the total number of unique values is limited. Fourth, we also show an adversarial perspective when the training data are in the same domain as IID but can lead to low reliability. This means that CNNs, when used in visualizations, can easily be fooled (section 6). Finally, CNNs have much lower errors and uncertainty than humans in some of the testing conditions (section 7). These results emphasize the importance of formalizing the training sampling domain of quantitative data when evaluating CNNs in visualizations.

Our work makes the following **contributions**: (1) A fundamental understanding of CNNs’ behaviors by focusing on *quantitative* data sampling unique to the visualization context; (2) A reusable controlled study method for comparing observers, either humans or CNNs; (3) New knowledge about CNNs’ behaviors through a set of experiments to quantify aspects of train-test discrepancies. We hope our research will both inspire new systematic AI behavior experiments in visualizations and support a cost-effective interpretation strategy to understand complex algorithmic behaviors.

2 BACKGROUND AND RELATED WORK

This section reviews the work that inspired ours: chart perception experiments in visualizations and quantification studies of CNNs.

2.1 Sampling Methods for Quantifying Visual Perception

Sampling is crucial in comparing and assessing the responses of both algorithmic and human observers, particularly in visualization tasks such as chart reading. In human studies, controlled sampling helps isolate the factors that influence reading. Cleveland and McGill in their 1984 seminal work (C-M) [8] sampled 7 out of 9 ratios (1st row in Figure 2) derived from 10 heights: 10, 12.1, 14.7, 17.8, 21.5, 26.1, 31.6, 38.3, 46.4, 56.2, resulting in spaced ratio samples seen by human observers. These samples were spaced evenly on a logarithmic scale to represent the power-law distribution of human length perception [27]. This work has been replicated and extended to crowdsourcing studies by Heer and Bostock [22], who chose a slightly different set of ratios (2nd row in Figure 2) from C-M’s original selection. Talbot et al. [49] adopted

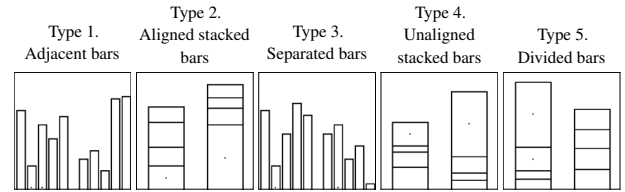


Figure 3: Our study reused the five bar types from C-M [8].

C-M’s ratios but instead sampled bar heights (3rd row in Figure 2) corresponding to those ratios.

Progress toward machine systems that quantify human-like understanding of visualizations has incorporated training data from human studies. For example, Haehn et al. [20] rounded C-M’s original 10 heights to the nearest integers, yielding 43 unique ratios (last row in Figure 2), which were randomly allocated for model training, validation, and test. This method assessed model inference accuracy under a specific human test data preparation, which did not consider that humans and models may have distinct decision mechanisms [16]. For example, the use of human samples to train and test CNNs could have conferred advantage to humans, since humans learn from a few samples and require fewer test instances to study decision mechanisms. In contrast, CNNs typically require exposure to a broader and denser sampling of the data domain to achieve comparable performance to human observers. The samples used in Haehn et al. represented only a proportion of available samples in the training and test sets.

In this work, we systematically controlled training and test data to assess algorithmic behaviors. Our present study further suggested that the sampling method in Haehn et al. [20] to compare humans and CNNs may also conferred an advantage to their CNNs, because the test and training samples are adjacent to each other (our main finding was that the performance is influenced by the distances). Our new sampling-based evaluation method allows us to draw new conclusions and offer deeper insight into when and why CNNs succeed.

2.2 Understanding Machine Learning Algorithm Behaviors

Our work is related to understanding the behavior of CNNs and aligns with three broad research directions. The first method concerns *explainable* approaches that introduce additional computational models [31] or visualizations [36, 47, 51, 58, 59] to interpret black-box behaviors. We avoided introducing new models, as this process itself is error-prone [45]. We also avoided visualization-based explainable methods (e.g., [6, 37, 53, 54]), as they are not quantitative [4] and may not be consistently perceived by human observers [17]. The second method involves human-algorithm alignment, where algorithms are used to explain human decision-making. This approach has only recently led to significant advances, offering a broad, global understanding of observers’ decisions [10, 39], primarily for categorical data in computer vision. The third method aims to trace algorithmic responses back to their training data by analyzing CNNs’ characteristics to interpret model behaviors [4, 15, 55, 57]. This method has also been only used in visualizations by posing curated questions to LLMs [3, 19, 25, 52]. However, these work focuses on mean accuracy alone, and none performed behavior analyses to explain the underlying mechanisms.

In our work, we adopt the third approach to examine CNNs’ behaviors by probing their outputs through controlled alterations in training distributions. By analyzing the outputs point-by-point in response to different training inputs, we aim to better understand how to train CNNs and when their inferences can be trusted. To our knowledge, this represents the first attribution study specifically designed to quantify CNN observers’ behaviors in data visualizations.

2.3 The Cost of Data Diversity

Producing large training sets is often slow, and the associated data wrangling, acquisition, and cleaning can also be prohibitively expensive [29]. Therefore, we examine the reliability of the model under resource constraints to evaluate how models perform with varying

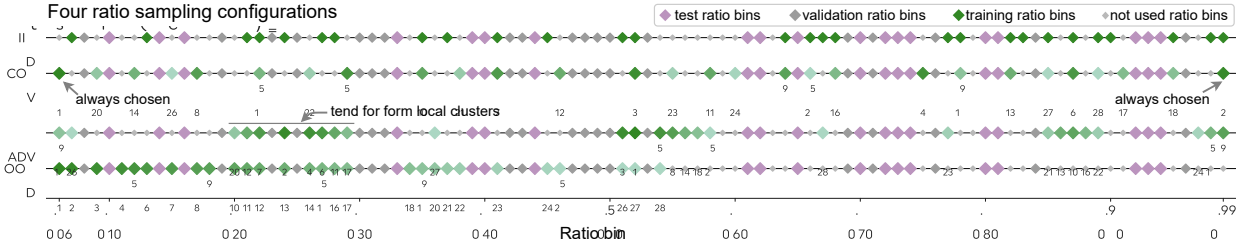


Figure 4: The ratio bin data sampling configurations in our sampling regime. The axis represents 94 unique ratio bins $\in [0.06, 0.99]$. The numbers below the training samples $\blacklozenge$ and their luminance show the order n which the samples are chosen for each of the sampling methods. The smaller the number or the darker the $\blacklozenge$ color, the earlier a ratio bin is chosen. **Observations.** For the training samples $\blacklozenge$, COV bins are spread out; ADV bins form clusters; OOD bins do not fully represent the test distribution. (Apdx. Figure 24 has the full list of all runs for ratio sampling.)

numbers of unique inputs. Furthermore, we investigated an adversarial condition. In vision science, this term is defined as “carefully crafted, seemingly normal inputs that have been slightly modified with imperceptible perturbations, causing the network to misclassify the image” [18, 30]. We define adversarial cases in visualization as the training samples that share the same data domain as test samples but may impair CNNs’ graphical perception ability. We leveraged an adversarial condition where the training samples are further from the *test* samples, which can limit the reliability of the models.

3 A SAMPLING REGIME: MEASURING MODEL RESPONSE TO TRAINING-TEST SAMPLING DISTRIBUTION VARIATIONS

This section presents the first contribution of this work: a training data sampling regime that quantifies model behavior under varying training-test discrepancies in the ratio or height data sampling domain.

3.1 The Problem Domain: Tasks and Visualization

We used a ratio estimation task, “judge what percent the smaller is of the larger” [8], because it required CNNs to reason about spatial relationships and had the highest inference errors among all tasks [20].

Ratio estimation task uses five bar chart types shown in Figure 3. Types 1–3 are bottom-aligned and thus judgment is by perceiving *position* along the common scale; and Types 4 and 5 must be compared by perceiving different *bar lengths*. Two target bars in each image are marked by a black dot and are adjacent to each other in all but Type 3. The ratio to be estimated is calculated as ratio = shorter (h)/taller bar height (H). In the human studies of position-length [8, 22, 49], people preferred position over length, and Types 1–3 outperformed Types 4–5.

3.2 The Target Data Sampling Domains

In our experiment, we used either the ratio or the taller bar height as the sampling target. Table 1 shows the range and count in the data domain.

Bar heights are integers: the taller bar height H ranges from 6 to 85 (80 values), and the shorter bar height h ranges from 5 to 84 (80 values), resulting in 3,240 unique (h, H) pairs. From these pairs, 2,081 unique ratios were derived, spanning from $5/85=0.059$ to $84/85=0.988$. We grouped these ratios into 94 evenly spaced bins (rb): $rb_1 : [0.055, 0.065)$, $rb_2 : [0.065, 0.075)$, ..., and $rb_{94} : [0.985, 0.995)$. For simplicity, each ratio bin is referred to by its midpoint value, such that $rb_1 : 0.06$, $rb_2 : 0.07$, ..., $rb_{94} : 0.99$, with $rb \in [0.06, 0.99]$. For Type 5, where two target bars are stacked on the same side, the constraint $h + H \leq 90$ reduced the sampling domain slightly, limiting the ratio bins to $rb \in [0.06, 0.98]$. Figure 4 and Figure 5 illustrate the ratio samples $\blacklozenge$ and height samples $\bullet$ along the axis, respectively.

3.3 Problem Framing and Sampling Methods

Framing ratio estimation task for CNNs. We frame the ratio estimation task as training a predictive model F that can take an input image x and output a predicted ratio $\hat{y}$.

Table 1: Data range and the total number of unique samples.

Ratio bin range (number of ratio bins)	Types 1-4: [0.06, 0.99]	(94)
	Type 5: [0.06, 0.98]	(93)
Taller bar height range (number of height samples)	Types 1-5 [6, 85]	(80)

Design principles. We assessed a CNN’s behavior by its responses to different sampling inputs by clearly defining the data domain. Given that model observers were compared in our experiments, we ensured a fair comparison by configuring conditions so that all observers took the same tests. In other words, test and validation sets were randomly chosen and held constant across different training conditions to ensure that models were evaluated using the same dataset for comparability. This control is crucial, as Cleveland and McGill [8] observed in their position-length experiment that human performance differed when judging ratios in different ratio bins. In our study, we also kept image appearances constant (See Apdx. Figure 26 for line width and gaps between bars). We detail below using ratio sampling as an example.

Step 1: Test and Validation Samples. We selected $94 \times 20\% = 19$ ratio bins for each of the test $\blacklozenge$ and validation $\blacklozenge$ sets.

Step 2: Selection of Training Data Samples. We assembled four sampling methods, each subsampled $m = 28$ ratios from 56 ratio bins, i.e., half of the remaining 60% ratio bins. Here, m could be set to any other numerical proportion of 56. We chose it for simplicity.

- **IID:** The *baseline* sampling method used to train IID. Training $\blacklozenge$ and test $\blacklozenge$ samples follow the same simple random sampling.
- **COV:** The *distant-to-train* condition used to prepare data sequentially by selecting $rb \blacklozenge$ that was the most distant from those that had been chosen so far. Thus, the first two rb selected were rb_1 and rb_{94} , the minimum and maximum of available ratio bins. The third rb was the most distant ratio bin from rb_1 and rb_{94} , and was close to the center of the sampling domain. This process continued, until $m = 28$ ratio bins were selected. In a sense, this sampling method encouraged the model to utilize the full data domain whenever possible.
- **ADV:** The *distant-to-test* condition used to prepare training samples $\blacklozenge$ as distant from the *test* samples $\blacklozenge$ as possible. This is aimed to evaluate an adversarial condition, where the chosen samples have the same domain as IID, thus *looking similar*, but were distanced themselves from the test set. We can see that these samples $\blacklozenge$ form local clusters, as adjacent points had similar distances to the test set.
- **OOD:** The *out-of-distribution* condition used to train OOD. We sorted all available 56 ratio bins in ascending order, and used the smallest 28 ratio bins to assemble a case of a biased training set. We called this OOD because the training set shifted to a different distribution from the test.

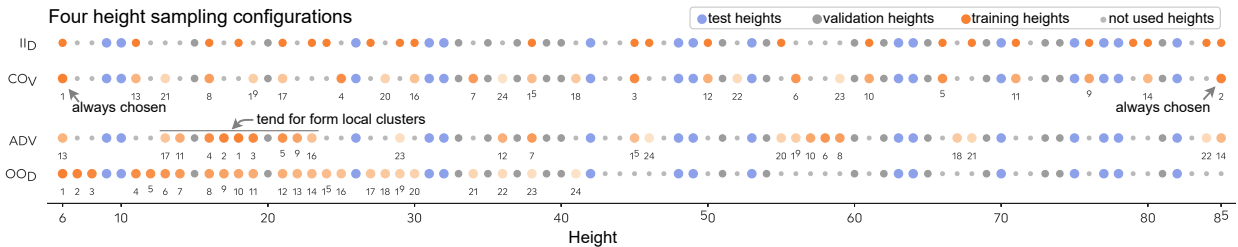


Figure 5: **The bar height data sampling configurations in our sampling regime.** The axis represents 80 unique taller bar heights $\in [6, 85]$. The numbers below the training samples $\bullet$ and the shading show the order in which the samples are chosen for each of the sampling methods. The smaller the number (or the darker the $\bullet$ color), the earlier a height is chosen. **Observations.** For the training set $\bullet$, COV samples are spread out. ADV tends to form local clusters, and OOD partially represents the test distribution. (Apdx. Figure 24 has the full list of all runs for height sampling.)

Step 3: Behaviors Under Limited Resources. Given this sampling configuration, we evaluated *stability under limited resources* by reducing the number of training samples by half each time. Starting with 30% of available ratio bins for training, we reduced this proportion to 15%, 7.5%, and 3.75%, resulting in 14, 7, or 3 ratio bins, following the labeled order along the axis in Figure 4.

Rendering Images. To generate an image, we first randomly select an rb in the training/validation/test set and then randomly select a (h, H) pair that yields a ratio within the selected bin, where h and H correspond to the shorter and taller target bar heights, respectively. For example, if $rb = 0.42$ was selected, both (19, 45) and (16, 38) are valid options for rendering the image. The heights of other non-target bars in the image are assigned randomly. See Apdx. Figure 22 for the full set of (h, H) pairs for each ratio bin and Figure 27 for examples of rendered images. We generated 60k/20k/20k images for training, validation, and test sets, respectively.

Model Configuration. We used VGG19 [46] and ResNet50 [21], conducting five runs for each sampling configuration to simulate the process as if five individuals were making multiple independent decisions. The images and their corresponding ground truth ratios served as inputs to supervise the CNNs. Since the test set remained consistent across sampling methods, the model inferences were directly comparable. (See Apdx. C for detailed model hyperparameter configurations.)

4 SUMMARY OF STUDIES

The following three sections detail our experimental methodology and findings. The studies were also registered at osf.io/gfqc3.

- **Study I: Sampling effect (section 5):** This study answers the question: *How sensitive are CNNs to out-of-distribution samples?* We controlled the sampling method, bar chart type, and CNN model architecture, evaluating behaviors using the mean absolute errors. Our findings indicate that CNNs could achieve considerable accuracy, but were not robust to the distribution shift. Additionally, CNNs' preference between position and length was minimal.
- **Study II: Stability (section 6):** This study answers the question: *How resilient are CNNs to limited input resources?* We controlled the sampling method and downsampling level, assessing models based on mean absolute error and models' intra-consistency. We found that downsampling the training set degraded model inference outcomes, and COV showed unexpected advantages that led to higher accuracy and reliability than IID.
- **Study III: Human-AI comparisons (section 7):** This study answers the question: *Can CNNs outperform humans?* We controlled the observer (humans vs. CNNs) and bar chart type. Model performance was assessed using the midmean log absolute error for comparisons with Cleveland and McGill [8] and the mean absolute error otherwise. We also measured the human-CNN inter-consistency. Results showed that CNNs had much lower errors and uncertainty than humans, and the errors of CNNs were also different from humans'.

5 STUDY I: MODEL ROBUSTNESS TO DISTRIBUTION SHIFT

This section reports the second contribution of this paper: answering *how models respond when training and test sets belong to different distributions*. We conducted two experiments, each involving either ratio or bar height sampling, by varying three independent variables one at a time: the sampling method (4 levels), the visualization type (5 levels), and CNNs (2 levels) in a full-factorial design. The configuration led to $4 \times 5 \times 2 \times 5 = 200$ runs in each experiment.

5.1 Design

5.1.1 Training Data Sampling

Exp 1 (ratio sampling robustness) sampled the ratio bins as described in subsection 3.3, illustrated in Figure 4.

Exp 2 (height sampling robustness) applied the same sampling process to the 80 unique heights (Figure 5), where 24, 16, and 16 unique heights were in the training, validation, and test sets, respectively.

5.1.2 Model Robustness Hypotheses

We had the following working hypotheses:

- **Study I: Sampling effect.H1 (Accuracy and Adversarial hypotheses).** Models will achieve high accuracy with IID and COV sampling, but not with ADV and OOD sampling.
- **Study I: Sampling effect.H2 (Out-of-distribution hypothesis).** Models trained in OOD will have reduced accuracy in regions where the training and test sets do not overlap.
- **Study I: Sampling effect.H3 (Chart type hypothesis):** We will not observe the effect of visualization types on inference accuracy.

Several reasons led to these hypotheses. Both IID and COV encouraged diversity, with IID being commonly used due to its ability to produce accurate inferences. COV also increased diversity by enforcing a greater separation between training samples. We did not expect ADV and OOD to perform well on average, as ADV represented an adversarial case, and OOD involved a distribution shift in the sampling domain. For the same reason, in H2, we did not expect accuracy gains for test values that were absent from the training data domain. H3 could be supported because CNNs were largely affected by data sampling input, rather than human perception limits.

5.2 Analysis Method

Each CNN model produced 20K inferences. Given the large number of inferences, we made several decisions when analyzing the data. First, we aggregated predictions by (h, H) pairs, treating each aggregation as an independent trial for downstream statistical analysis. This was necessary to avoid false discoveries resulting from the large number of predictions. Second, the error metric was mean absolute error (MAE), defined as $MAE = |\text{model inference } \hat{y} - \text{ground truth } y|$, consistent with the evaluation metric used in human observer experiments by Talbot et al. [49]. We did not use the logarithmic mean as in the C-M study [8] in this and the next sections, because taking the logarithm could disproportionately emphasize smaller errors. We also did not

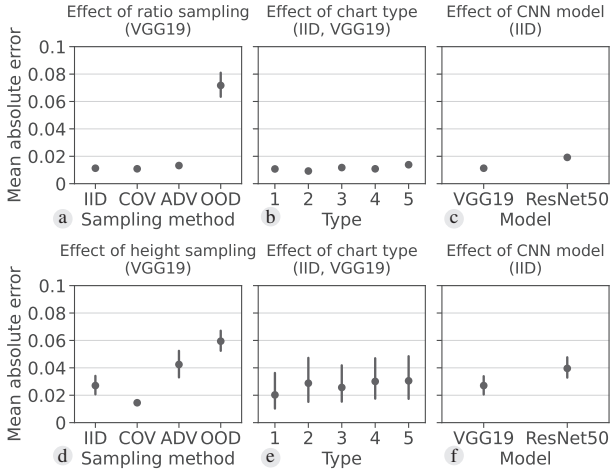


Figure 6: **Study I: Sampling effect.** Effect of sampling method, chart type, and CNN model from ratio sampling (a - c) and height sampling experiments (d - f). Error bars show 95% confidence intervals. **Observations.** (1) a and d, OOD had much larger mean inference errors than other sampling methods. (2) b and e, chart type differences were not dominant. (3) c and f, VGG19 had smaller errors than ResNet50.

Table 2: **Study I: Sampling effect.summary statistics.** $a > b$ means a is statistically more accurate than b . Medium effect size ($\eta^2 \geq 0.06$) is in *italic*, and the large one ($\eta^2 \geq 0.14$) is in **bold**.

Variable	F	p	η^2	HSD
Study I Exp 1: Ratio sampling				
Sampling (VGG19)	$F_{(3,2048)} = 174.6$	< 0.001	0.20	(COV, IID, ADV) > (OOD)
Type (IID, VGG19)	$F_{(4,509)} = 9.0$	< 0.001	<i>0.07</i>	(1, 2, 4) > (1, 3, 4) > (3, 5)
Model (IID)	$F_{(1,1022)} = 337.1$	< 0.001	0.23	(VGG19) > (ResNet50)
Study I Exp 2: Height sampling				
Sampling (VGG19)	$F_{(3,1591)} = 29.4$	< 0.001	<i>0.05</i>	(COV, IID) > (ADV) > (OOD)
Type (IID, VGG19)	$F_{(4,395)} = 0.3$	0.882	< 0.01	
Model (IID)	$F_{(1,794)} = 5.7$	0.017	< 0.01	(VGG19) > (ResNet50)

use mid-mean to trim 50% data in these first two studies for practical considerations: we were interested in CNNs' behaviors in real-world uses, where determining when to trim data may not always be feasible.

We examined the main effect of independent variables on prediction errors and performed statistical analysis using the General Linear Model (GLM). When a significant main effect was observed, we performed a post-hoc analysis using Tukey's Honest Significant Difference (HSD) test to assess the significance of differences across the levels in a treatment condition. Finally, we calculated the effect size using Cohen's η^2 [9] to measure the practical significance: "no effect" ($\eta^2 < 0.01$), "small effect" ($\eta^2 \in [0.01, 0.06]$), "medium effect" ($\eta^2 \in [0.06, 0.14]$), and "large effect" ($\eta^2 \geq 0.14$).

5.3 Results

Overview. We collected a total of 4 million predictions from 200 models in each experiment (200 models $\times$ 20K predictions). Figure 6 and Table 2 show the mean errors and the summary statistics.

Sampling method is a significant main effect. Study I: Sampling effect.H1 was supported for height sampling but not for the ratio sampling experiment. For the ratio sampling experiment, ADV (0.013) was in the same Tukey group as COV (0.011) and IID (0.011) (Figure 6 a), meaning that we did not observe a statistical difference among them. One reason for the lack of significance for ADV could be that it contained sufficient unique samples across the data domain.

For height sampling, COV (0.015) and IID (0.027) had lower errors than ADV (0.042) and OOD (0.059) (Figure 6 d). Now ADV was in a different Tukey group (Table 2). We suspected that the differences between IID, COV, and ADV conditions were caused by errors from

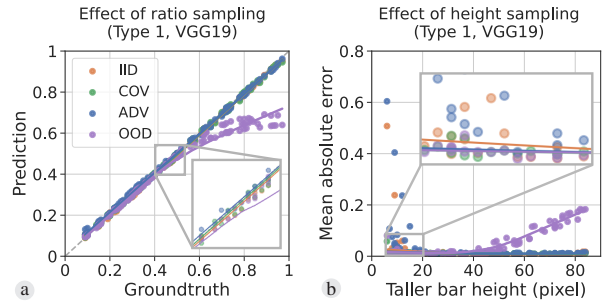


Figure 7: **Study I: Sampling effect.** Behavior analysis for Type 1 charts running VGG19. Regression lines show locally weighted scatterplot smoothing (LOWESS) [7]. **Observations.** a, OOD was not robust to extrapolation in both experiments. b, in the height sampling experiment, small bar heights led to a significant increase in inference errors.

those smaller bar heights (Figure 7) which we analyzed next.

Striking generalization failure on OOD. Study I: Sampling effect.H2 was supported. To study the CNNs' behaviors, we analyzed case-by-case CNNs' inference accuracy by ratio bins (Figure 7 a) and by heights (Figure 7 b) computed from the two experiments respectively. In Figure 7 a, the diagonal line means that the prediction is the ground truth, and the closer to this line, the smaller the inference errors. Figure 7 b shows the mean absolute errors against bar heights, lower error indicates higher inference accuracy. We observe that knee-shaped curves emerge in the out-of-train regions in OOD in both ratio sampling and height sampling experiments, showing errors that increase almost linearly. Interestingly, the in-train test data samples in OOD maintain low errors (0.008 for ratio sampling and 0.025 for height sampling). This result may support that CNNs' inference accuracy depends on the coverage of training input: IID did not guarantee sampling of the smallest or the largest values compared to COV and the minimum values of OOD. Our experiments in the next section validate this conjecture by stress-testing CNNs with limited training samples.

CNNs had limited ability to read smaller bar heights. We performed Tukey's HSD test on the 80 bar height levels found that smaller bar heights ($H \leq 15$ pixels) increased inference errors, independent of sampling methods. This small height effect was also observed in the ratio sampling experiment, where having $H \leq 25$ pixels increased the inference errors (See Apdx. Figure 29).

The effect of types varied by experiments. Sampling effect H3 was partially supported. Type was a significant main effect in ratio sampling only. We see that CNNs' differences in inference accuracies between types are much smaller than those of humans in C-M [8].

Model was also a significant main effect. Specifically, VGG19 outperformed ResNet50 in our experiments. We suspect that this performance difference can be attributed primarily to the way we configured the network, rather than to any inherent advantages or disadvantages of these architectures. Given these findings, we focus our subsequent discussion on results obtained for Type 1 and VGG19, and leave other types and ResNet50 results in Apdx. Figures 28—33 and Table 6.

6 STUDY II: DOWNSAMPLING TO TEST MODEL STABILITY

This section reports results of the *sampling stability*, which examines how the number of unique training samples impacts inference errors. Answers to this question can help reduce training costs and guide the selection of most resilient sampling strategies. We also ran two full-factorial experiments on ratio and bar height, with two independent variables—the sampling method (4-levels) and downsampling level (3-levels). Each configuration was repeated 5 times, resulting in a total of

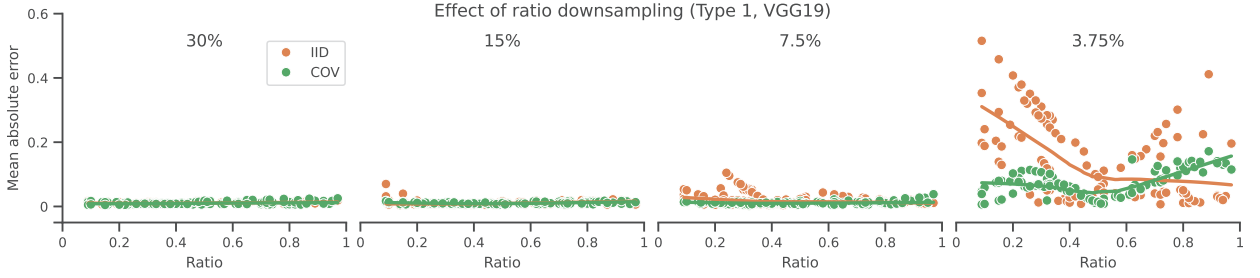


Figure 8: **Study II: Stability.** *Ratio downsampling case-by-case analysis for Type 1, VGG19 at four downsampling levels.* **Observations.** As the number of training samples decreases, COV exhibited stronger stability compared to IID. While IID initially perform similarly to COV, its errors increased substantially under further downsampling. This was due to the formation of small clusters, causing many test samples to fall out-of-distribution and leading IID to behave more like OOD.

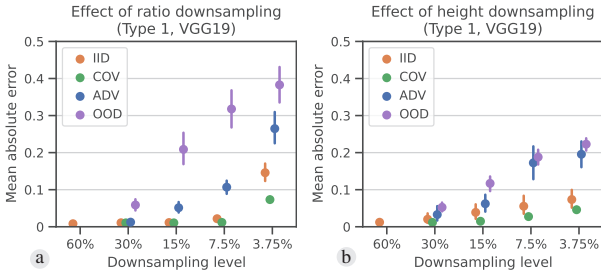


Figure 9: **Study II: Stability.** *Effect of downsampling on model stability for a ratio and b height sampling experiments.* **Observations.** (1) COV was most resilient to downsampling. (2) ADV was not stable and its errors increased as quickly as OOD.

$4 \times 3 \times 5 = 60$ models per experiment. In addition to evaluating *accuracy*, we measured *intra-consistency* to capture the variability in model inferences for a given sampling method across different downsampling levels. Higher consistency indicates greater model stability.

6.1 Design

6.1.1 Training Data Sampling

We used the same test sets as Study I. The implementation of downsampling was straightforward, described in section 3.3. The training samples were a subset of those used in Study I, section 5. Three downsampling levels of training were used: ratio bins of 14, 7, or 3, and height levels of 12, 6, or 3, corresponding to downsampling rates of 15%, 7.5%, or 3.75%, respectively, as shown in our sampling configurations in Figure 4 and Figure 5.

6.1.2 Model Stability Hypotheses

We had the following working hypotheses:

- **Study II: Stability.H1** (*downsampling accuracy hypothesis*): Decreasing the number of unique samples in the training set will increase CNNs' inference errors.
- **Study II: Stability.H2** (*downsampling stability hypothesis*): The four sampling methods will not be equally robust.

These hypotheses could be supported because COV and IID remained diverse, while OOD could introduce greater distribution shifts towards the smaller values. Under constrained resources, ADV also introduced out-of-distribution cases, further reducing its reliability.

6.2 Analysis Method

We used the same method to analyze inference errors as described in Study I. In addition to mean absolute error, we further measured the

Table 3: **Study II: Stability.** Summary statistics.

Variable	F	p	η^2	HSD
Study II Exp 1: Ratio downsampling				
Level	$F_{(3,1641)} = 150.2$	< 0.001	0.17	(30%) > (15%) > (7.5%) > (3.75%)
15%	$F_{(3,408)} = 68.2$	< 0.001	0.33	(COV, IID, ADV) > (OOD)
7.5%	$F_{(3,408)} = 112.1$	< 0.001	0.45	(COV, IID) > (ADV) > (OOD)
3.75%	$F_{(3,408)} = 56.6$	< 0.001	0.29	(COV) > (IID) > (ADV) > (OOD)
Study II Exp 2: Height downsampling				
Level	$F_{(3,1273)} = 70.6$	< 0.001	0.12	(30%) > (15%) > (7.5%) > (3.75%)
15%	$F_{(3,316)} = 21.6$	< 0.001	0.17	(COV, IID) > (IID, ADV) > (OOD)
7.5%	$F_{(3,316)} = 34.1$	< 0.001	0.24	(COV, IID) > (OOD, ADV)
3.75%	$F_{(3,316)} = 46.7$	< 0.001	0.31	(COV, IID) > (OOD, ADV)

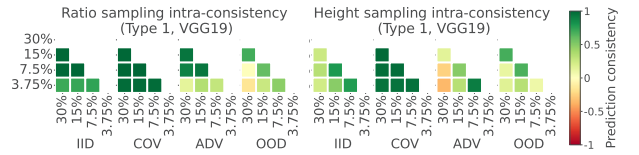


Figure 10: **Study II: Stability.** *Intra-consistency computes the correlation of model inferences between different downsampling levels.* **Observations.** COV had the highest consistency for both ratio and height sampling experiments.

intra-consistency of model inferences between downsampling levels by the Pearson correlation coefficient [9]. A higher Pearson correlation indicates greater intra-consistency and stability for a sampling method in response to limited resources.

6.3 Results

Overview. We collected a total of 1.2 million predictions (60 models $\times$ 20K predictions) for each of the ratio and height stability experiments. Figure 9 and Table 3 show the mean errors and the summary statistics. Both hypotheses were supported. (More analyses in Apdx. Figure 30.)

Reducing the number of unique training samples increased inference errors, with COV exhibiting half the errors of IID. Study II: Stability.H1 was supported. The downsampling level was a significant main effect on errors. In the ratio sampling experiment, inference errors for IID increased more than tenfold, from 0.011 at a downsampling level of 30% to 0.146 at 3.75%. For COV, errors rose about sevenfold, from 0.010 to 0.073. COV and IID were in the same Tukey group until the 3.75% downsampling condition (Table 3). At 15% downsampling level, ADV (0.051) was in the same group as IID (0.011) and COV (0.010), but its error increased sharply ($> 10\times$) between 7.5% and 3.75%. These results suggested that COV was the most stable sampling method among the four. The same trend was observed in the bar height experiment, where COV again produced the lowest errors (Figure 9).

COV is the most stable sampling method. Stability.H2 was sup-

ported. We assessed model stability using intra-consistency, calculated as the correlation of model inferences across different downsampling levels. As shown in Figure 10, IID and COV had the largest intra-consistency ($r > 0.99$) between the 30%, 15%, and 7.5% levels in the ratio downsampling experiment. For IID, consistency dropped to $r = 0.72$ when including the 3.75% level, whereas COV maintained a high consistency of $r = 0.98$. Intra-consistencies were much lower for ADV ($r = 0.59$) and OOD ($r = 0.32$) on average. In the height downsampling experiment, COV also showed the highest intra-consistency ($r = 0.97$), compared to IID ($r = 0.48$). ADV ($r = 0.21$) had even lower consistency than OOD ($r = 0.34$). These results indicate that the COV sampling method is the most stable across different downsampling levels, maintaining high intra-consistency even at extreme reductions.

Figure 8 provides a case-by-case analysis of the ratio downsampling experiment, further illustrating the stability of COV. OOD errors increased almost linearly as test cases moved further out-of-distribution. Both IID and ADV also exhibited OOD-like behavior when training samples were limited. In contrast, COV enforced in-distribution conditions for all test cases, preventing errors from rising linearly; instead, they remained bounded by the evenly distributed training samples. This is most evident at level 3.75%, where only three samples were used for training. The errors of COV were stable, and lower errors occurred when test samples were close to the training samples—specifically, at the leftmost, rightmost, and middle positions.

7 STUDY III: HUMAN VS. CNN: A COMPARATIVE STUDY

Given the relatively small errors in the first two studies when applying IID and COV, we proceeded with a rigorous human-CNN comparison, extending our preliminary findings [28] to answer the question: *to what extent do CNNs and humans differ?* For the human observer, the bar chart type was the only independent variable considered. For the CNN observer, we introduced a new baseline condition (IID-large) that utilized the entire remaining 60% of ratio bins or heights without downsampling. This corresponded to 56 ratio bins in the ratio sampling experiment or 48 height bins in the height sampling experiment. We also reused the IID and OOD conditions from Study I section 5. We did not use COV since its error was bounded by the error of IID.

7.1 Design

7.1.1 Human Experiment Data Collection

We adapted the position-length experiment in Cleveland and McGill [8] using the 10 integer bar heights of 10, 12, 15, 18, 22, 26, 32, 38, 46, 56. Exhausting all pairs generated 45 height combinations, resulting in 37 unique ratios across 23 ratio bins: 0.18, 0.26, 0.27, 0.38, 0.39, 0.45, 0.46, 0.47, 0.48, 0.55, 0.56, 0.57, 0.58, 0.67, 0.68, 0.69, 0.7, 0.8, 0.81, 0.82, 0.83, 0.84, 0.85.

We conducted the human experiment on the LabInTheWild [44] crowdsourcing platform. We did not use the Amazon Mechanical Turk, because our pilot study had much higher mean errors on that platform compared to the results presented by Cleveland and McGill [8]. In contrast, data collected via LabInTheWild had demonstrated quality comparable to the C-M experiment (See the experiment user interface in Apdx. Figure 31 and platform comparisons in Apdx. Figure 32). Participants had the same instruction as that of C-M study, to make a quick division between the heights of the shorter and the taller marked bars. They first completed 5 training trials—one for each bar chart type—with immediate feedback provided after each response. After that, each participant completed 25 trials, consisting of 5 chart types with 5 trials per type in random order.

7.1.2 Human-AI Comparison Hypothesis

We had the following working hypotheses:

- **Study III: Human-AI comparisons.H1 (accuracy hypothesis):** Our CNNs' errors will be significantly lower than humans'.

This hypothesis could be supported based on our observation comparing our results in the first two experiments to those in the human experiments of Cleveland and McGill [8]. CNNs appeared to have much smaller errors compared to those of human observers.

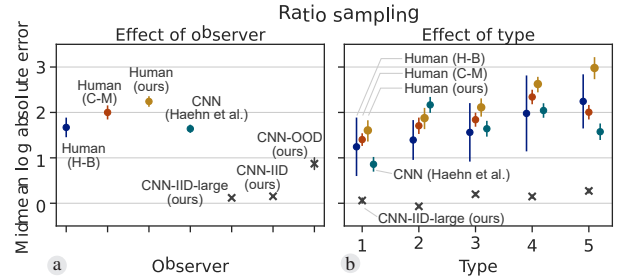


Figure 11: **Study III: Human-AI comparisons. Ratio sampling.** **a** Comparison of the average errors across five bar chart types for seven observers: (1) Cleveland and McGill (C-M) [8], (2) Heer and Bostock (H-B) [22], (3) Haehn et al. [20], (4) our human experiment, and (5)-(7) our three CNNs (IID-large, IID, and OOD). **b** Per-type comparison among observers. Error bars are 95% confidence intervals. **Observations.** (1) From **a**, our human results are comparable to those of C-M [8] and Heer and Bostock [22]. Our CNNs lowered errors greatly compared to that of Haehn et al. [20]. (2) From **b**, CNNs didn't have strong preference for certain types, unlike humans.

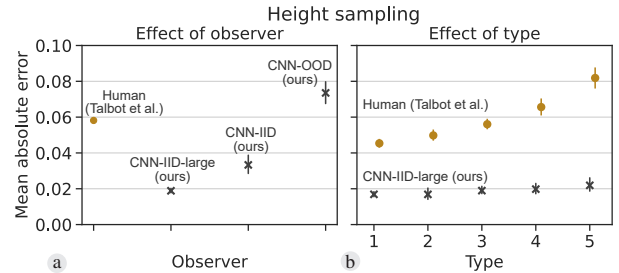


Figure 12: **Study III: Human-AI comparisons. Height sampling.** Mean absolute errors between Talbot et al. [49] and our CNNs. Error bars are 95% confidence intervals. **Observations.** (1) From **a** CNNs outperformed humans for all but OOD-sampling conditions. (2) From **b** chart type had more influence on humans than on CNNs.

Table 4: **Study III: Human-AI comparisons. Summary statistics.**

Variable	F	p	η^2	HSD
Study Ixv Exp 1: Ratio sampling				
Observer	$F_{(3,3327)} = 334.3$	< 0.001	0.22	(IID-large, IID) > (OOD) > (Human)
Type (human)	$F_{(4,249)} = 24.6$	< 0.001	0.28	(1, 2, 4) > (1, 3, 4) > (3, 4, 5)
Study Ixv Exp 2: Height sampling				
Observer	$F_{(3,7278)} = 151.2$	< 0.001	0.06	(IID-large) > (IID) > (Human) > (OOD)
Type (human)	$F_{(4,4882)} = 50.9$	< 0.001	0.04	(1, 2) > (2, 3) > (3, 4) > (5)

7.2 Analysis Method

We used mid-means of the log absolute error (MLAE) [8] to measure accuracy, in order to compare our results to those published in literature, i. e., Cleveland and McGill [8], Heer and Bostock [22], and Haehn et al. [20]. Here $MLAE = \log_2(|\text{model inference} - \text{ground truth}| + 1/8)$. The height sampling experiments used mean absolute error to compare to that of Talbot et al. [49]. Finally, we evaluated human-CNN inter-consistency using Pearson correlation analyses.

7.3 Results

Overview. We collected human observers' results from 1,575 trials from 63 participants, who spent an average of 10.0 seconds per trial. Figure 11, Figure 12, and Table 4 show the mean errors and the summary statistics. Overall, our human experiment results showed slightly higher error than Cleveland and McGill's (Figure 11 a), but the trend across the five chart types remained consistent with the findings of Cleveland and McGill [8] and Heer and Bostock [22] (Figure 11 b).

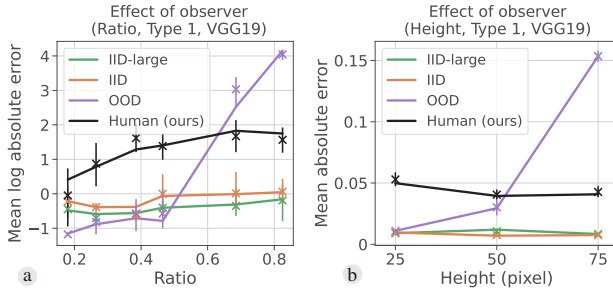


Figure 13: **Study III: Human-AI comparisons.** Case-by-case analysis of inference errors between humans and CNNs. **Observations.** CNNs can outperform humans in general, but if not trained properly, they may perform worse than humans on out-of-distribution tests.

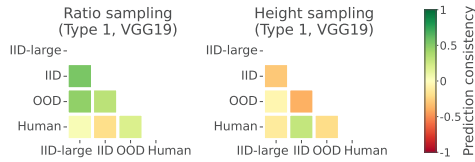


Figure 14: **Study III: Human-AI comparisons.** Inter-consistency between humans and CNNs. **Observations.** The inter-consistency between humans and CNNs is relatively low.

(See Apdx. Figure 32 for comparisons between our replication and Cleveland and McGill’s original results).

CNNs had lower errors than humans. Study III: Human-AI comparisons. H1 was supported. The observer was a main effect: human errors were much higher than those of CNNs. For ratio sampling, human errors were 4 times higher IID-large for ratio sampling (0.056 vs. 0.013), Figure 11 a), and for height sampling, human errors (0.058) were about three times that of CNNs’ IID-large (0.019) (Figure 12 a). However, CNNs are not robust to out-of-distribution tests (Figure 13).

Chart type was also a significant main effect. The differences between types were observed in human experiments (Figure 11 b and Figure 12 b). In addition, CNNs had smaller uncertainty (95% CI for Type 1, Baseline, ratio: 0.002, height: 0.002) compared to human observers (95% CI, ratio: 0.007, height: 0.004).

Human-CNN inter-consistency is lower than the intra-consistency between CNNs (Figure 14). In ratio sampling: $r = 0.01, p > 0.1$; and in height sampling: $r = -0.02, p > 0.1$. This suggested that human errors differed from those in CNNs.

8 GENERAL DISCUSSION

This section presents explanations for our findings, reflects on approaches for quantifying CNNs’ capabilities, and provides practical recommendations for training data preparation.

8.1 The Effect of Training-Test Distances and Reusability

One of the most intriguing findings in our study was the stability of the COV sampling method, particularly when the total number of training samples was limited (Figure 9). We hypothesize that this stability stems from the key property of COV: it ensures a broad coverage of the data space while maintaining an even distribution of training samples. By maintaining a well-spread distribution of training samples, COV effectively minimizes large gaps in the training space, ensuring that test samples are likely to have nearby training samples for reference.

To validate this hypothesis, we analyzed the relationship between inference error and the training-test distance—the average distance from each test sample to its closest training sample. As shown in Figure 15, there was a strong positive correlation in both ratio ($r = 0.98$) and height ($r = 0.94$) sampling experiments, confirming that models performed more accurately when test samples were closer to their nearest training neighbors. Among all sampling strategies evaluated,

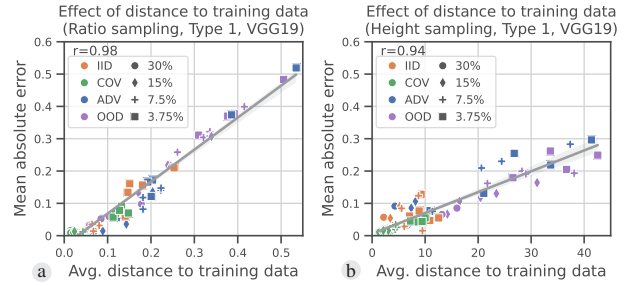


Figure 15: **Inference errors vs. training-test distance.** Each point represents a run and there were 80 runs (5 runs $\times$ 4 sampling methods $\times$ 4 downsampling levels). **Observations.** We observed a strong correlation ($r = 0.98$ and 0.94) between the training-test distance and the inference error. Compared to other sampling methods, COV $\bullet$ tends to distribute samples more evenly, resulting in a smaller training-test distance and smaller inference errors.

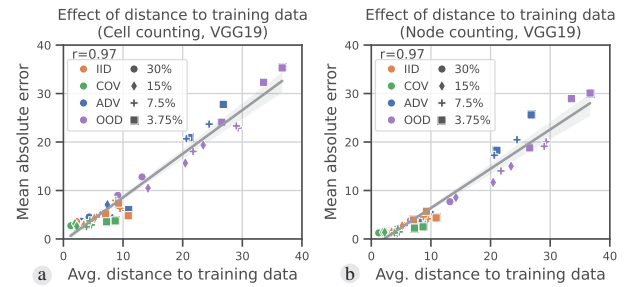


Figure 16: **Reuse of our stability results for counting tasks.** Inference errors vs. training-test distance for cell and node counting tasks. **Observations.** The same strong correlations between the training-test distance and the model inference error were found in both experiments (cell: $r = 0.97$, node: $r = 0.97$), suggesting that the effect of training-test distance can be generalized. Apdx. subsection D.3 has study configurations.

COV had the lowest training-test distance at equivalent downsampling levels, further supporting its advantage in stability.

To examine the reusability of this stability result, we investigated whether the impact of training-test distance extends beyond the ratio estimation task. We observed a similar pattern in two additional challenging real-world counting tasks. The first is counting cells in fluorescence-light microscopy images rendered using the method of Lehmussola et al. [33]. It required CNNs to identify cells based on continuous color, textures, and shapes. The second is counting network nodes in node-link diagrams rendered using high-fidelity graph simulations by Lancichinetti et al. [32]. We manipulated the community size and required CNNs to perceive the shape of the diagram to make a prediction. The results in Figure 16 suggest that the effect of training-test distance can be generalized to new tasks.

8.2 Implications to Training Data Preparation Concerning Reliability, Stability, and Cost-effectiveness

The results of our study suggest that the training sampling choices were a significant main effect on CNNs’ inference errors. Using a few human ratio samples may disadvantage CNNs, as exposure to a wider range of ratio samples could improve their capabilities. Implicitly mixed sampling conditions in the training could potentially increase inference uncertainty, because inference errors from an OOD-trained model could be much larger than those from an IID-trained model. On the other hand, conditions that benefit CNNs could also appear. When training and test samples are close neighbors, the learning algorithm can gain an advantage, leading to higher accuracy.

From a **reliability** perspective, we found that CNNs could achieve considerable accuracy. However, they are not robust to the train-test distribution shift. In-distribution tests remained accurate, but errors

increased linearly for out-of-distribution cases (Figure 7). From a **stability** perspective, models should try to cover the data range as much as possible, especially when resources are limited or when sampling more data is too expensive (Figure 15). Under limited resources, the most commonly used IID sampling method does not guarantee stability (Figure 10). The lack of robustness in the ADV condition also suggests that chart reading CNNs could also be easily fooled in visualization uses. From a **cost-effective** perspective, we found that using COV can substantially reduce the training time: on average 69.7%, 82.6%, and 96.4% less time compared to IID, ADV, and OOD, respectively (Apdx. Figure 37).

Furthermore, our current experiment only tested only a single **adversarial** case, in which training and test samples share the same domain but are distant from each other, a scenario that can severely degrade the accuracy of CNNs. It would be valuable in future work to consider attacks that also affect human observers, such as those presented by Correll and Heer and Correll et al. [11, 12]. Finally, we suggest enhancing transparency to mitigate misuse and misinterpretation. A practical measure is to incorporate a model card [40], providing information on how training samples are selected, along with their ranges and distributions, image size, and the number of unique samples.

8.3 Comparison to the Results of Haehn et al. [20]

Haehn et al. [20] stated, “In other experiments, such as the position-length experiment, CNNs cannot complete the task.” However, our Study III in section 7 demonstrated that CNNs can outperform humans. We provide an explanation for this mismatch from a data sampling perspective. In our experimental setup, the heights of target bar samples ranged in [5, 85] pixels, whereas Haehn et al. followed Cleveland and McGill’s approach, using only 10 discrete heights. As a result, our configuration led to a substantially larger set of unique (h, H) pairs (see Apdx. Figure 22). Although the smaller training-test distance in Haehn et al.’s experiment (as shown in Figure 2) provided great advantages to CNN observers, their limited variation in training (h, H) pairs was among the reasons that hindered models’ inference accuracy.

From the attribution study perspectives, their approach of random assignment of the C-M test samples into the training, validation, and test sets made it hard to analyze the influence of the training set on large errors and uncertainty in model inference outcomes. Their analyses may have introduced large variations in test conditions (see Apdx. Figure 23). In contrast, our study explicitly controlled training and test sets, providing an interpretable perspective.

These sampling methodological differences underscore the significant impact of data preparation strategies on CNNs’ behaviors. Our findings are consistent with recent research showing that the behavior constraints of CNNs differ from those of humans [16]. Directly adopting experimental designs from human studies may introduce methodological biases, potentially resulting in models that do not provide fair comparisons between observers [14]. As such, comparing models to human observers must carefully state sampling regimes, and an absence of a fixed, identical test set may not constitute a fair evaluation.

8.4 The effect of Small Stimuli

Our findings from Studies I and II showed that small bar heights (≤ 15 pixels for height sampling and ≤ 25 pixels for ratio sampling) increased errors for IID and ADV, but not for OOD (Figure 7). The first explanation was that OOD encountered more images with short bars during training. A second explanation could be the effect of sidedness: we only selected the left side with small bar heights in our experiments and did not test OOD using regions with the largest bar heights. If similar errors occurred in the larger-side sampling, it would suggest that the error was not due to bar height itself but rather a consequence of our design choice. We thus examined the other side of the bar height or ratio samples, to train models. We observed an asymmetry in the height sampling conditions, and saw that smaller heights indeed led to larger errors (Figure 17).

This behavior contrasts with human perception that follows Weber’s Law [27], making ratio judgments independent of the absolute magnitude of the stimuli. Apdx. subsection D.1 provides explanations to the

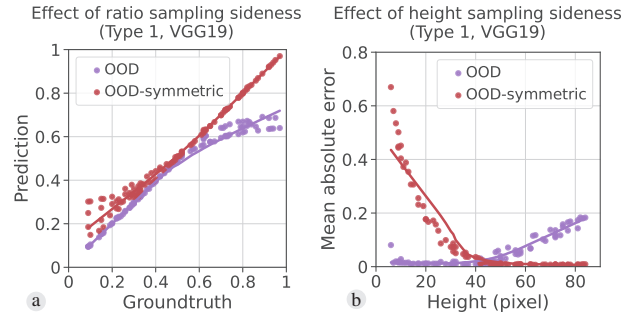


Figure 17: **Impact of smaller bar heights on OOD.** Comparison of inference errors between OOD and OOD-symmetric (opposite direction sampling). **Observations.** Data sampling symmetric conditions (OOD vs. OOD-symmetric) exhibited asymmetric errors in b height sampling only.

short height effect from CNNs’ architectural perspective. To improve this condition, one can increase the input image size so that smaller bar heights are proportionally increased, or use networks and modules that are designed for small-size elements [34, 35].

8.5 Limitations and Future Work

In this work, we designed a new attribution study method to quantify algorithmic behaviors by explicitly considering training-test discrepancies, offering foundational insights into effective data sampling for visualization tasks. While our study focuses on four sampling strategies, other sophisticated approaches, such as those that follow a target data distribution [1], may prove more effective and are worthy of future investigation. Also, addressing the intricate contexts and patterns of practical analytical tasks will require future studies to effectively leverage our quantification methods. Finally, fundamental theories for quantifying CNNs remain scarce. Methods to test the generalizability of the findings to other visualization types, beyond those tested here, are needed to study the alignment between humans and machines for visualization applications.

9 CONCLUSIONS

Our sampling regime to quantifying CNNs reveals CNNs’ mechanism to chart reading suggests the need for rigorous training-test separation to study CNNs in visualization uses. Coverage-based sampling (COV) emerged as a robust strategy, more robust than the mostly used IID, largely due to its maximized data span and evenly distributed samples. Not surprisingly, while models achieve superhuman accuracy in controlled settings, inference errors increase for out-of-train-data samples (distribution shifts). Lastly, smaller visual stimulus (e.g., short bars) introduces larger inference errors, suggesting that training must account for visual mark size.

Our research findings have also yielded new insights into methods to quantify CNNs’ behaviors to interpret charts.

- **Quantification Method:** Reporting case-by-case accuracy and beyond-average performance metrics is more informative. Notably, even in OOD conditions, in-sample data still performed well.
- **Efficiency:** Prioritize diverse sampling over large datasets—COV requires 50% fewer samples than IID for comparable accuracy.
- **Transparency:** When deploying CNNs in real-world visualization tasks, model cards should include details on sampling methods, such as data span and train-test distributions. This helps us know when to trust AIs and why not to.
- **Visual encoding:** Sufficiently large visual marks may be necessary to reduce errors. Otherwise, increase the number of unique training samples is necessary.

Beyond the recommendations we have made, we hope our sampling regime and the probing methods provide the community with a means to quantify behavior differences between humans and machines.

ACKNOWLEDGMENTS

We thank Justin Talbot, Vidya Setlur, and Anushka Anand for sharing their results, which we have used for human-CNN comparisons in [section 7](#). We also thank all participants who volunteered in our online experiment for their time and effort. H. Pfister was partially supported by NSF grant CRCNS-2309041. Finally, we thank the reviewers for their feedback, which helped us improve the quality of this paper.

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